

Speech Tech Week 3

1) Which of the following sets of parameters uniquely define a Gaussian distribution?

1 point

- ☐ μ
- ☐ σ
- ☒ μ, σ
- ☐ None of the above

$$N(\mu, \sigma)$$

2) By changing the parameter _____ the height of the bell curve can be changed.

1 point

- ☐ mean
- ☐ mean or standard deviation
- ☒ standard deviation or variance
- ☐ mean or variance

→ std. dev.

peak → mean
distance → std dev.

1) Probability can be calculated from the probability distribution function (PDF) by calculating:

1 point

- ☒ Area of Probability Distribution Function
- ☐ Height of Probability Distribution Function
- ☐ none of the above
- ☐ Both A and B



→ area under the curve
→ if enemy not present but my sys. says so.

2) Given the radar system, which of the following is false alarm?

1 point

- ☒ Enemy target is not present and the decision from Gaussian distribution estimates that it's present
- ☐ Enemy target is present and the decision from Gaussian distribution estimates that it's present
- ☐ Enemy target is present and the decision from Gaussian distribution estimates that it's not present
- ☐ none of the above

1) If you are given height and weight of all the citizens in a country and you are asked to model this data, which of the following will you use?

1 point

- ☐ uni-variate Gaussian
- ☒ bi-variate Gaussian
- ☐ None of the above

→ 2 variables

2) In the case of bi-variate Gaussian, mean is a _____

1 point

- ☐ scalar
- ☒ vector

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

↓
vector

$$\sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$$

↓
matrix

1) Which of the following sets of parameters uniquely define a Gaussian Mixture Model (GMM)?

1 point

- ☐ μ (Mean)
- ☐ σ^2 (Variance)
- ☐ π (mixture probability)
- ☐ μ, σ^2

μ, σ, π

2) Estimating the parameters of a GMM is a/an?

1 point

- ☐ Supervised Learning Problem
- ☒ Unsupervised Learning Problem
- ☐ Semi-supervised Learning Problem
- ☐ None of the above

we do it through an iterative process until convergence

3) Given a GMM density function $f(x)$. Which of the following is true:

1 point

- ☐ $\int_{-\infty}^{\infty} f(x) dx < 1$
- ☐ $\int_{-\infty}^{\infty} f(x) dx > 0$
- ☐ $0 < \int_{-\infty}^{\infty} f(x) dx < 1$
- ☒ $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\int f(x) dx = 1$$

1) Which of the following is the probability density function (pdf) $f(x)$ of the Gaussian Mixture Model with k mixtures?

1 point

- ☐ $\sum_{i=1}^k \pi_i N(x, \mu_i, \sigma_i^2)$ where $\sum_{i=1}^k \pi_i < 1$
- ☐ $\sum_{i=1}^k \pi_i N(x, \mu_i, \sigma_i^2)$ where $\sum_{i=1}^k \pi_i > 0$
- ☒ $\sum_{i=1}^k \pi_i N(x, \mu_i, \sigma_i^2)$
- ☐ None of the above

2) Number of components in a Gaussian Mixture Model is learned while estimating the parameters

1 point

- ☐ True
- ☒ False

false, it is known already

3) In GMM, each data point may belong to :

1 point

- ☐ Only a single mixture
- ☐ At most, two mixtures
- ☒ Multiple mixtures, each with respective probability
- ☐ None of these

1) In EM Algorithm, During E Step, we estimate mean (μ), variance (σ^2), and mixture probability (π) using pseudo labels.

1 point

- ☐ True
☒ False

True

2) In EM Algorithm, During M Step, we estimate mean (μ), variance (σ^2), and mixture probability (π) using pseudo labels.

1 point

- ☒ True
☐ False

3) Latent variables are also called :

1 point

- ☐ pseudo label
☐ hidden variables
☒ both A and B
☐ None of these

4) Before we use EM-Algorithm, we randomly initialize the parameters of a GMM

1 point

- ☒ True
☐ False

1) Binary Representations is an example of Whole number quantization

1 point

- ☒ True
☐ False

2) Which of the following function is an example of Real number quantization

1 point

- ☐ Floor
☐ Ceil
☒ Both A and B
☐ None of these

1) By changing the parameter _____, we can change the position of the peak of the bell curve or Gaussian curve.

mean

1 point

2) By what method, the Mean and standard deviation of Gaussian curve are estimated?

maximum likelihood estimation

1 point

3) Given a GMM density function $f(x)$. Which of the following is true:

1 point

- ☐ $\int_{-\infty}^{\infty} f(x) dx < 1$
☐ $\int_{-\infty}^{\infty} f(x) dx > 0$
☐ $0 < \int_{-\infty}^{\infty} f(x) dx < 1$
☒ $\int_{-\infty}^{\infty} f(x) dx = 1$

4) Which of the following is the probability density function (pdf) $f(x)$ of the Gaussian Mixture Model with k mixtures?

1 point

- ☐ $\sum_{i=1}^k \pi_i N(x, \mu_i, \sigma_i^2)$ where $\sum_{i=1}^k \pi_i < 1$
☐ $\sum_{i=1}^k \pi_i N(x, \mu_i, \sigma_i^2)$ where $\sum_{i=1}^k \pi_i > 0$
☒ $\sum_{i=1}^k \pi_i N(x, \mu_i, \sigma_i^2)$ where $\sum_{i=1}^k \pi_i = 1$
☐ None of the above

5) We use Maximum Likelihood Estimation to estimate the mean and variance for each component

1 point

☒ True

☐ False

6) Which of the following statements are true about GMM:

1 point

☐ GMM only allows hard assignments of input data points to the mixtures

☐ GMM only allows soft assignments of input data points to the mixtures

☒ GMM allows both hard and soft assignments of input data points to the mixtures

☐ None of these

both true labels and pseudo labels

7) EM Algorithm will always lead to a global maximum solution

1 point

☐ True

☒ False

convergence for sure

8) Which of the following is true about the EM-Algorithm :

1 point

☐ EM-Algorithm does not converge

☐ EM-Algorithm uses Maximum Likelihood Estimation to estimate parameters.

☐ EM-Algorithm is an Iterative Algorithm

☒ both B and C

1) Which of the following is the probability density function (pdf) of uni-variate Gaussian distribution?

1 point

☐ $-\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2}$

☒ $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2}\right)$

☐ $\frac{1}{\sqrt{2\sigma\pi}} \exp\left(-\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2}\right)$

☐ None of the above

Yes, the answer is correct.

Score: 1

Accepted Answers:

$$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2}\right)$$

2) In the classification of Male and female height scenario, for which case the decision will be Male in the following options:

1 point

☒ $f_{\text{male}}(h_0) dx \geq f_{\text{female}}(h_0) dx$

☐ $f_{\text{male}}(h_0) dx \leq f_{\text{female}}(h_0) dx$

☐ $f_{\text{male}}(h_0) dx = f_{\text{female}}(h_0) dx$

☐ None of the above

Yes, the answer is correct.

Score: 1

3) Number of parameters in a k mixture uni-variate GMM are:

1 point

- ☐ $2 \times k$
- ☒ $3 \times k$
- ☐ $2 \times k - 1$
- ☐ $2 \times k + 1$

$$k \Rightarrow 3k$$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$3 \times k$

4) Which of the following statement of the Gaussian Mixture model is NOT correct?

1 point

- ☒ Gaussian mixture model can be viewed as a machine from which a data point is generated using one of the Gaussian distributions with equal probability.
- ☐ Given the set of the training data points, the Gaussian mixture model can be learned by using an Expectation Maximization algorithm to maximize the likelihood of generating these data from the model.
- ☐ Once the model is learned and the parameters are learned, any new data point can be classified into one of the Gaussian components with the maximum probability.
- ☒ None of these

No, the answer is incorrect.

Score: 0

Accepted Answers:

None of these

5) We have 100 data points representing the height of the population containing Adults and Children. Out of 100 data points, 30 data points represent the height of Adults. We want to fit a GMM with two mixtures where mixture probability for Adults and Children are given by π_A and π_C , respectively. π_A and π_C will be :

1 point

- ☐ 0.5, 0.5
- ☐ 0.7, 0.3
- ☒ 0.3, 0.7
- ☐ None of these

$$\pi_A = \frac{30}{100} = 0.3$$

Yes, the answer is correct.

Score: 1

Accepted Answers:

0.3, 0.7

$$\pi_C = 1 - 0.3 = 0.7$$

6) For a given set of data points $X = \{x_1, x_2, \dots, x_N\}$, log-likelihood for a GMM with k mixtures can be defined as:

1 point

- ☐ $\sum_{j=1}^N \log(\sum_{i=1}^k \pi_i N(x_j, \mu_i, \sigma_i^2))$
- ☐ $\log(\prod_{j=1}^N \sum_{i=1}^k \pi_i N(x_j, \mu_i, \sigma_i^2))$

☒ Both A and B

☐ None of these

Yes, the answer is correct.

Score: 1

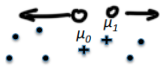
Accepted Answers:

both A and B

$\log(\pi) \Rightarrow$ leads to summan of log

$E \rightarrow$ computes responsibilities (soft assign. n)
 $M \rightarrow$ moves each mean to the weighted average of the data assigned to

7) Suppose you are applying EM (Expectation-Maximisation) to train a Gaussian Mixture Model to cluster the data below into two clusters. The + points indicate the current means μ_0 and μ_1 of the two Gaussian mixture components after the k^{th} iteration of EM. What will be the direction in which μ_0 and μ_1 will move during the M step? 1 point



that comp

- ☒ μ_0 moves to the left, and μ_1 moves to the right
- ☐ μ_0 moves to the right, and μ_1 moves to the left
- ☐ μ_0 moves to the right, and μ_1 moves to the right
- ☐ μ_0 moves to the left, and μ_1 moves to the left

Yes, the answer is correct.

Score: 1

Accepted Answers:

μ_0 moves to the left, and μ_1 moves to the right

8) Which of the following statements are true regarding vector quantization 1 point

- ☒ Idea behind vector quantization is to divide a large set of vectors into a small group of vectors
- ☐ K-means is not an example of vector quantization

☒ Vector quantization is used to compress information

☒ Both A and C

it is means of clustering algo.